Prashant Sah

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Education

Caldwell University **Caldwell, NJ**
Major: Computer Science, **Double Minor:** Mathematics, Business Analytics **May 2027**

- Dean's List, Honors Award Recipient, Presidential Scholarship
- Courses: Data Structures & Algorithms, Web Development, Database Management, Calculus, Artificial Intelligence

Projects

Next Word Predictor | PyTorch, LSTM, NLTK GitHub

- Designed and implemented an LSTM-based next-word prediction model using PyTorch with an embedding → LSTM → linear architecture.
- Built a complete NLP pipeline including NLTK tokenization, vocabulary construction (249 tokens), sequence generation (631 sequences), and padded batching (max length: 127).
- Trained the model for 50 epochs (batch size: 32) using Adam optimizer and cross-entropy loss, achieving **99.52% next-word prediction accuracy** with final loss of 1.35.
- Implemented both single-step and iterative text generation for autocomplete-style sequence completion.

Video Content Retrieval-Augmented Generation (RAG) System GitHub

- Engineered end-to-end machine learning pipelines using Python, pandas, and scikit-learn for automated data preprocessing, feature extraction, and model training.
- Implemented natural language processing (NLP) workflows integrating OpenAI Whisper and BGE-M3 embeddings for semantic text analysis and context understanding.
- Developed retrieval-augmented generation (RAG) models leveraging vector embeddings and cosine similarity for context-aware information retrieval.
- Automated data ingestion, transformation, and parallel processing for JSON and multimedia datasets using pandas, NumPy, and Joblib to optimize pipeline efficiency.

California Housing Price Predictor GitHub

- Achieved 97.1% predictive accuracy (R-squared) using Random Forest regression with cross-validation scores of 96.6% ± 0.6% across 5-fold validation, demonstrating robust model performance.
- Built Flask web application with 9-feature ML pipeline including automated data preprocessing, handling 1,000+ housing records with stratified sampling for unbiased model training
- Deployed production-ready application on Render with RESTful API endpoints, supporting both single predictions and batch processing of CSV files up to 16MB
- Implemented automated CI/CD pipeline with GitHub integration, reducing deployment time from manual setup to 6-11 minutes with automatic model retraining on code updates

Experience

Caldwell University, Teaching Assistant – CS 196 & CS 302 **Jan 2025 – Dec 2025**

- Assisted in instruction, assignment design, and grading for 50+ students in Python programming, covering object-oriented programming (OOP), recursion, data structures, and data manipulation.
- Conducted hands-on lab sessions on data analysis and visualization using NumPy, Pandas, Matplotlib, and Seaborn, improving students' practical coding and analytical skills.
- Provided academic support for algorithms coursework (15 students), including algorithm design, graph theory, complexity analysis, debugging, and code optimization, enhancing code efficiency and problem-solving accuracy.

Caldwell University, Student Tech – IT Department **July 2024 – Present**

- Supported campus-wide AV and IT systems, ensuring reliable data and technology operations.
- Assisted faculty and students with troubleshooting university portal and system access issues.
- Rebuilt and configured computers, optimizing system performance and data processing efficiency.
- Collaborated with IT staff to maintain hardware, software, and network integrity for data-driven tasks.

Caldwell University, Peer Tutor – Academic Success Center **Sept 2024 – Present**

- Tutored 25+ students in Python and algorithms, strengthening their computational problem-solving skills.
- Guided learners through topics in web development, discrete math, and algorithmic logic.
- Assisted students in applying programming concepts to data analysis and efficient code design.

Caldwell University, Barnes & Noble Campus Store Team Member **July 2024 – Present**

- Analyzed textbook inventory, enrollment data, and sales trends to forecast demand and maintain accurate stock levels.
- Used POS and inventory systems to ensure accurate data entry and support data-driven decisions.

Skills

Languages: Python, JavaScript, C, SQL (MySQL), HTML/CSS
Libraries & Frameworks: Pandas, NumPy, Scikit-learn, Matplotlib, Flask, PyTorch
Tools: Git/GitHub, Jupyter Notebook, VS Code, Excel, Word
Domains: Machine Learning, Data Analysis, NLP, Data Visualization